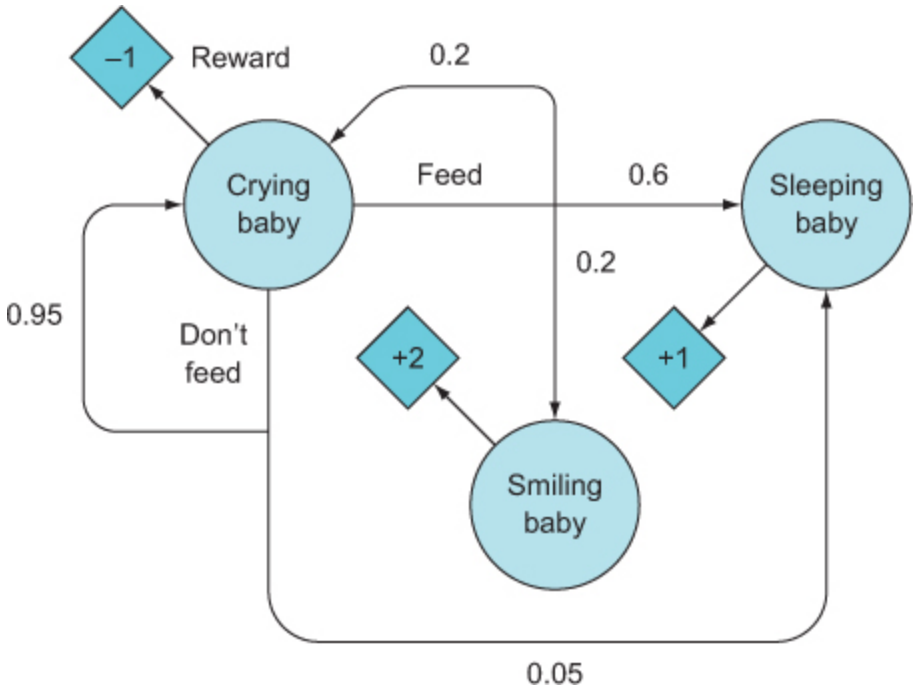


2.7. Predicting future rewards: Value and policy functions

Believe it or not, we actually smuggled a lot of knowledge into the previous sections. The way we set up our solutions to the *n*-armed bandit and contextual bandit are standard reinforcement learning methods, and as such, there is a whole bunch of established terminology and mathematics behind what we did. We introduced a few terms already, such as state and action spaces, but we mostly just described things in natural language. In order for you to understand the latest RL research papers and for us to make future chapters less verbose, it's important to become acquainted with the jargon and the mathematics.

Let's review and formalize what you've learned so far (summarized in figure 2.9). A reinforcement learning algorithm essentially constructs an *agent*, which acts in some *environment*. The environment is often a game, but is more generally whatever process produces states, actions, and rewards. The agent has access to the current state of the environment, which is all the data about the environment at a particular time point, $s_t \in S$. Using this state information, the agent takes an action, $a_t \in A$, which may